

Mahmoud Youssef

Backend Engineer | Java · Spring Boot · Redis · MySQL

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SUMMARY

Backend Engineer specializing in Java and Spring Boot, focused on production-style systems with authentication, caching, and distributed-system patterns (locking, idempotency, sharding). Designs, builds, tests, and deploys backend systems independently, from data modeling through deployment.

EDUCATION

El Shorouk Academy <i>Bachelor of Science (B.Sc.) in Computer Science</i>	Cairo, Egypt Jul 2026
Harvard University (via edX / ITI) <i>CS50x — Introduction to Computer Science</i>	Online 2024

PROJECTS

Attendance Management API <i>Spring Boot 3, MySQL, Redis, JWT, Docker</i>	GitHub
- Built a 34-endpoint attendance system with QR check-in, HMAC-SHA256 signed sessions, and Redis-backed per-employee deduplication to prevent duplicate check-ins across concurrent team sessions - Designed daily <code>@Scheduled</code> reconciliation to auto-mark absences and sync approved leave; implemented reporting via native SQL recursive CTEs, optimized to avoid N+1 query overhead	
LogiBridge — B2B Order Management Platform <i>Spring Boot 3, PostgreSQL, Redis, Docker</i>	GitHub
- Built a 15+ endpoint order-management API with multi-role RBAC, JWT + refresh token rotation with reuse detection, and an immutable audit trail - Enforced safe concurrent order-state transitions using a DB unique constraint with pessimistic/optimistic locking; added DB-backed idempotency keys for safe retries, validated with JUnit 5 + Mockito	
E-Commerce Backend System <i>Spring Boot 3, MySQL, Redis, Stripe, Docker</i>	GitHub
- Built a 30+ endpoint API with event-driven order/payment flow (<code>OrderCreatedEvent</code>, <code>PaymentCompletedEvent</code>), applying SOLID and Clean Architecture across modules - Implemented idempotent Stripe webhooks, two-phase inventory reservation with optimistic locking, and Redis cache-aside that cut read latency ~40%	
CalCounter — AI Nutrition Tracker <i>Spring Boot 3, MySQL, JWT, HuggingFace, Docker</i>	GitHub · Demo
- Led backend development for a 3-person graduation team, building a 30+ endpoint JWT/RBAC-secured API deployed on Render with a Flutter web client on Firebase Hosting - Cut food-photo analysis wait time ~60% by integrating a HuggingFace EfficientNetB3 model through an async pipeline; beta-tested by 15+ users	
University Timetable Scheduler <i>Spring Boot 3, MySQL, Docker</i>	GitHub
- Built a full-stack scheduling system enforcing 7 hard and 5 soft constraints, with async generation, status polling, and JWT role-based access - Shipped a React/TypeScript SPA with an interactive weekly grid and PDF/Excel/PNG export; deployed via Docker Compose with GitHub Actions CI/CD	
Distributed URL Shortener <i>Spring Boot 3, MySQL, Docker</i>	GitHub
- Built collision-resistant ID generation with Redis INCR + Base62 encoding; added atomic per-IP rate limiting via Redis - Implemented deterministic hash-based sharding across MySQL shards with no lookup table, plus cache-aside reads and scheduled cleanup of expired records	

TECHNICAL SKILLS

Languages: Java (OOP, Collections, Streams, Lambda Expressions), SQL
Backend: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, MapStruct, JdbcTemplate
Databases: MySQL, PostgreSQL, Redis (caching, rate limiting, session/TTL state)
Security: JWT Authentication, Refresh Token Rotation, RBAC, HMAC-SHA256 Signing
DevOps & Tools: Docker, Git, GitHub Actions CI/CD, Maven, Flyway, Swagger/OpenAPI, Postman
Testing & Concepts: JUnit 5, Mockito, OOP, SOLID, Design Patterns, Clean Architecture, REST APIs